INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II , Second Semester Examination, March- 2025

Subject Code: BCACC201

Subject Name: Data Structures and Algorithms

Full Marks: 30 Set No: 3 Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. Define node in linked list. 2

2. Differentiate between pre-order and in-order traversal of a tree. 2

3. List the operation to be performed on a binary tree. 2

4. Time complexity of quick sort and bubble sort. 2

5. What are the different types of sorting algorithms? 2

6. What do you mean by output restricted queue? 2

Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. Describe the steps involved in reversing a singly linked list, and write the function for it. 5

8. Explain the insertion and deletion at front in dequeue. 5

9. Construct an expression tree for the expression (a+b*c) + ((d*e+f)*g). Give the outputs when you apply in-order and pre-order traversals. 5

Part - C
Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. Write a C program to implement queue using linked list. 10

11. Develop an algorithm or code to implement insertion sort. 10

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